More Nines in Deep RL Reliability

Workshop update: Project 15, Week 3

Speaker A | 0:00-0:40

GOAL

Make easy control tasks reliable, not just high return on average.

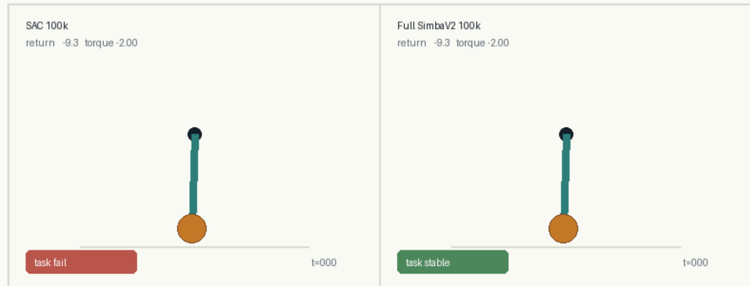
Pendulum should be close to solved. The interesting failures are the last starts where a neural SAC policy still does not swing up and stabilize.

83.0%
+/- 7.8 pp SE
SAC 100k: ref success within 5

91.8%
+/- 0.8 pp SE
SimbaV2 100k: ref success within 5

81.2%
+/- 8.9 pp SE
SAC 100k: task-stability predicate

91.5%
+/- 0.6 pp SE
SimbaV2 100k: task-stability predicate



GIF: exact grid start initial angle=-174.1 deg, angular velocity=-1.00; SAC seed0 fails, full SimbaV2 seed0 succeeds.

What Is Success?

Speaker A | 0:40-1:35

Why the headline metric uses max(DP, controller)

First attempt: intuitive task success

We tried a behavioral criterion: swing up and stay near upright.

- Near upright if `cos(theta) >= 0.95` and `|theta_dot| <= 1.0`.
- Stable if near-upright fraction is at least 80% during the 200-step rollout.
- No long loss if the longest not-near-upright streak is at most 50 steps.

Problem: not feasible everywhere

Our DP/controller references satisfy that task-stability predicate on **2336/2501** cells (93.4%). The other **165** cells are not certified by either reference, so task-stability alone is not a fair universal success definition.

Final headline success: match the best same-start reference

`R(policy, s0) >= max(R_DP(s0), R_controller(s0)) - 5`.

DP is the stronger return reference on **2403/2501** cells (96.1%); the hand controller is stronger on **98/2501** cells (3.9%). We use the max because DP is approximate and the controller fixes some regions.

Evaluation Protocol

What was run and how each plotted number is counted

Exact initial-state grid

- 61 angle bins x 41 velocity bins = **2501** reset-support states.
- Five training seeds per current 100k comparison, so **12505** deterministic grid rollouts.
- Map color means fraction of seeds satisfying the plotted criterion in that exact cell: 0%, 20%, 40%, 60%, 80%, 100%.
- Seeds 3-4 are follow-up runs with diagnostics every 10k steps instead of 100k; training recipe and final evaluation are otherwise the same.

Reference and diagnostics

- DP: finite-horizon value iteration on a 241 x 161 state grid and 81 torque actions, horizon 200.
- Controller: energy swing-up plus local PD stabilization.
- Trackers: replay near-upright coverage, critic dormancy, and effective rank.

What Did SimbaV2 Change?

Speaker A | 2:15-3:05

Four concrete changes relative to the CleanRL SAC baseline

1. Hyperspherical feature normalization

Replace layer norm-style scale learning with L2 normalization, so hidden feature vectors keep a fixed length instead of drifting to larger or smaller magnitudes.

2. Hyperspherical weight normalization

Remove weight decay and project selected weights back to unit norm after each gradient update. The optimizer mainly changes directions, not raw scale.

3. Distributional critic + reward scaling

Predict a distribution over return bins instead of one Q value. Scale rewards so the critic target stays near unit variance and gradient norms are less sensitive to outliers.

4. Official SimbaV2 SAC recipe

Use the paper-style optimizer and entropy settings rather than the default CleanRL SAC hyperparameters. This makes early learning slower but is meant to be more stable.

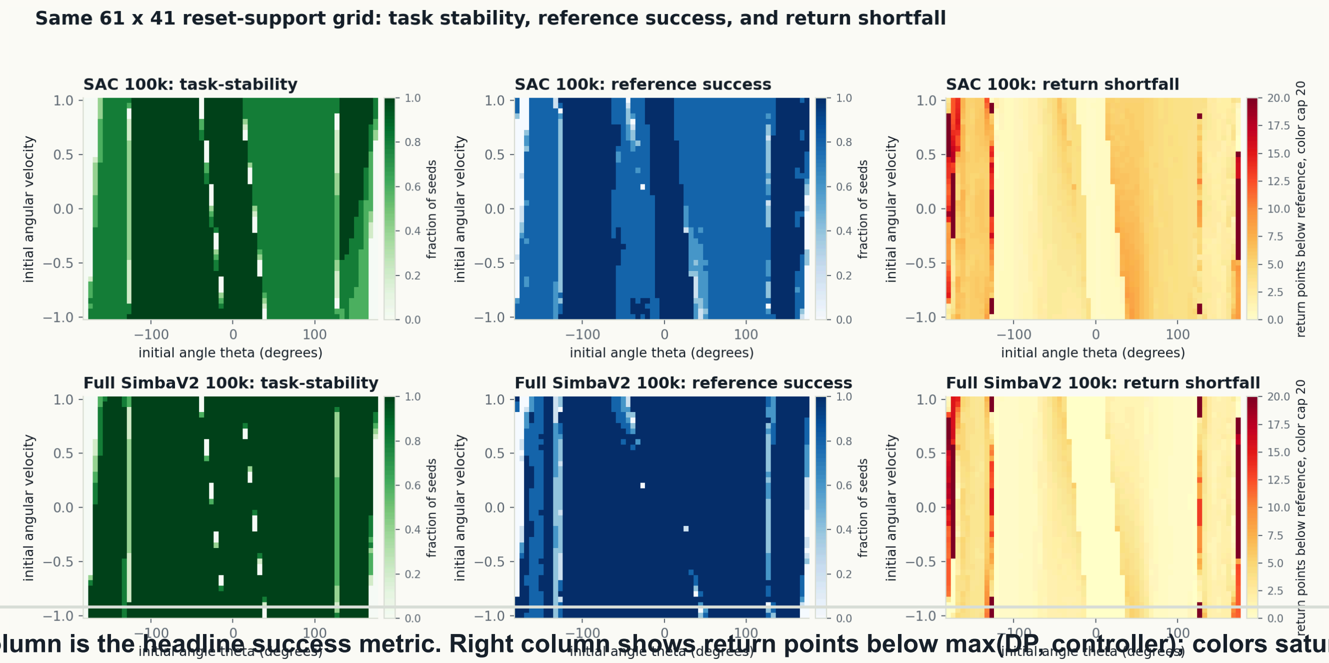
Parameter	CleanRL SAC baseline	Full SimbaV2 run
Q / actor learning rate	1e-3 / 3e-4	1e-4 -> 5e-5 / 1e-4 -> 5e-5
Initial entropy weight alpha	1.0	0.01
Target entropy scale	-1.0	-0.5
Network size	actor 128, critic 512	Simba actor 32, critic 64
Critic target	scalar Q value	51-bin distribution + reward scaling

We should not claim yet which of these four changes drives the 100k improvement. The component ablations are now a next experiment at representative budget.

Source: Lee et al., "Hyperspherical Normalization for Scalable Deep Reinforcement Learning", arXiv:2502.15280, 2025.

Raw Plots

The same starts under the two success criteria

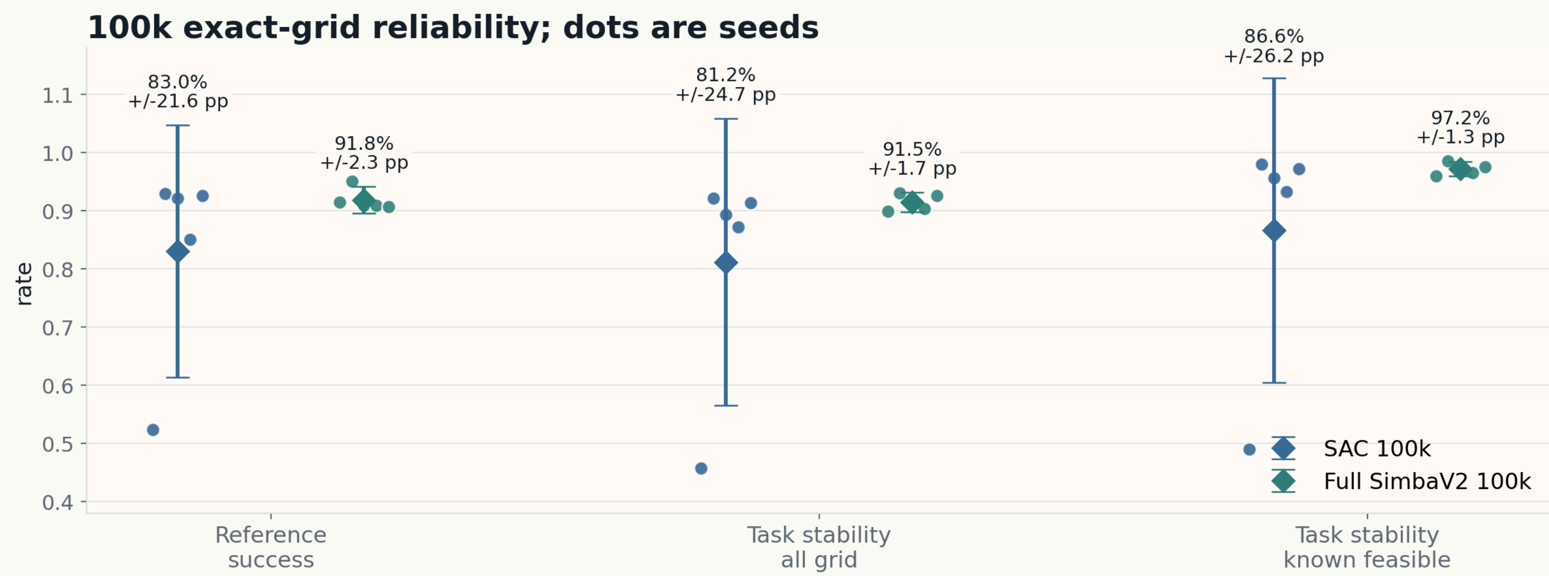


Middle column is the headline success metric. Right column shows return points below max(DP, controller); colors saturate at 20 because rare outliers exceed 100 and would wash out the map. Unclipped cell-mean shortfall: SAC max 122.4, mean 4.6 +/- 8.2; SimbaV2 max 113.8, mean 3.2 +/- 9.3.

First Result

Full SimbaV2 is the current 100k reference-success frontier

Speaker B | 3:50-4:35



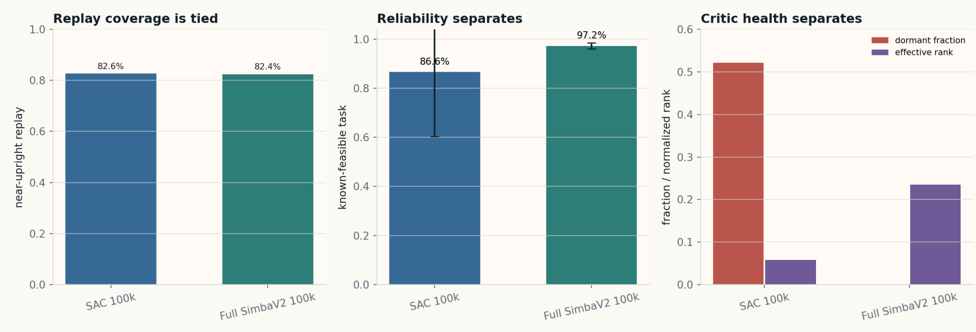
Diamonds are seed means; vertical intervals are 95% t-intervals over seeds (n=5).

Reference success: 83.0% +/- 21.6 pp for SAC vs 91.8% +/- 2.3 pp for SimbaV2. Task-stability all-grid: 81.2% +/- 24.7 pp vs 91.5% +/- 1.7 pp.

Exploration vs Optimization

Replay coverage is necessary, but not sufficient

Speaker B | 4:35-5:25



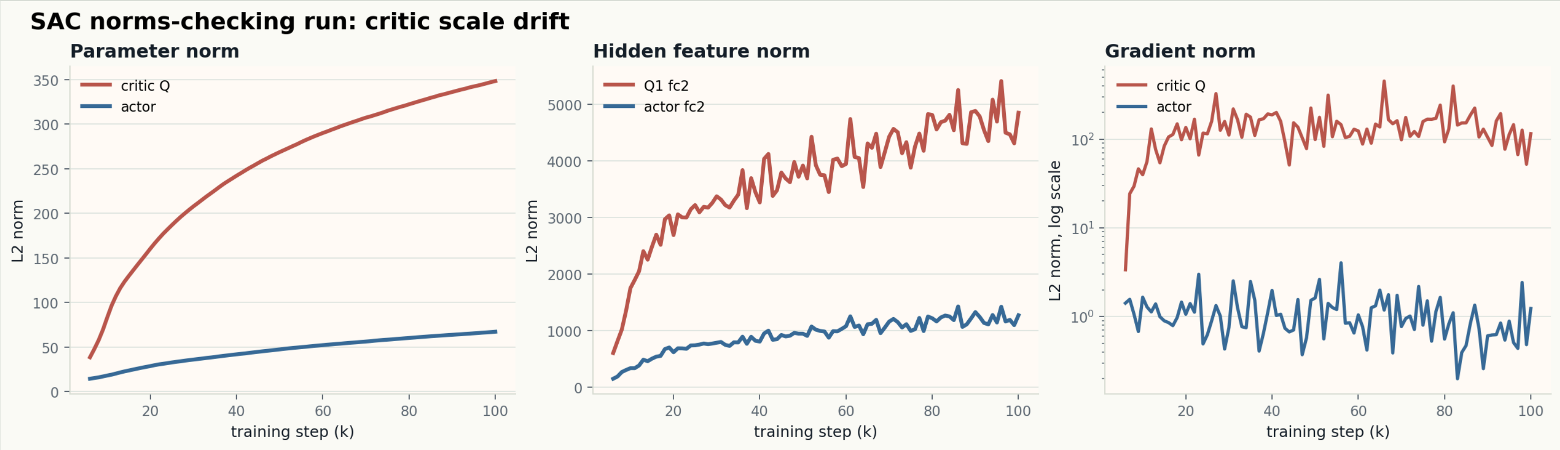
Current diagnosis

- Replay near-upright coverage: SAC 82.6%, SimbaV2 82.4%.
- Dormant critic units are units that barely activate; lower is better.
- Effective rank is a feature-diversity proxy; higher is better.
- Same replay coverage but different critic health points to optimization and plasticity, not pure exploration.

SAC Norm Diagnostics

A mechanism check for the optimization diagnosis

Speaker B | 5:25-5:55



Critic scale drifts

In the SAC norms-checking run, critic parameter norm grows from **38.4** to **348.5**; actor norm grows from **14.4** to **67.1**.

Features also inflate

Q1 fc2 activation norm grows from **606** to **4856** and peaks at **5416**.

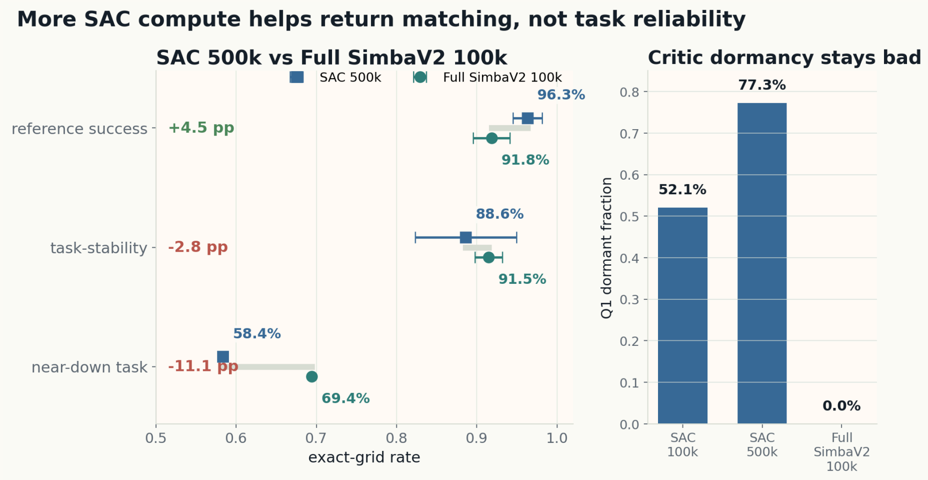
Scope of the claim

This is one SAC diagnostic run. It supports the critic-health story and motivates the 100k SimbaV2 component ablations.

What Did Not Solve It?

Speaker B | 5:55-6:30

More SAC compute improves return matching, but still misses task reliability



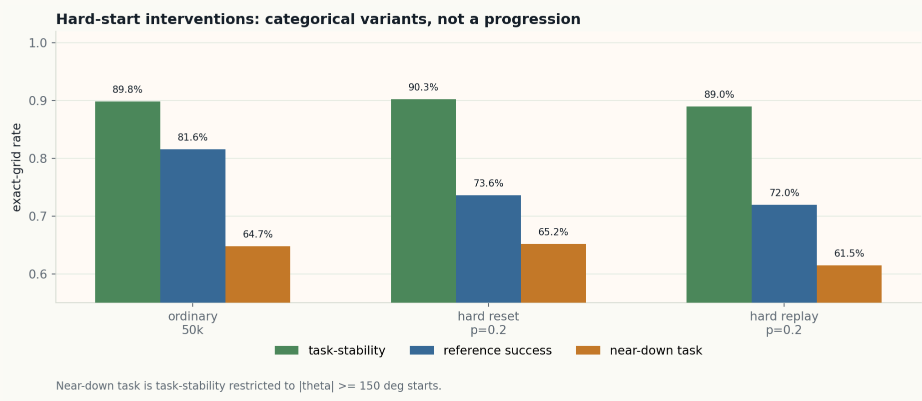
Why this is negative

- Reference success is not the problem: SAC 500k is 96.3% vs Full SimbaV2 100k at 91.8% (+4.5 pp).
- Task-stability is still worse: 88.6% vs 91.5% (-2.8 pp), and near-down task success is 58.4% vs 69.4% (-11.1 pp).
- Critic dormancy worsens with compute: SAC 500k has 77.3% dormant Q1 units vs 0.0% for Full SimbaV2 100k.

What Else Did Not Solve It?

Hard reset and hard replay are categorical data interventions

Speaker B | 6:30-7:10



Definitions

- Hard reset p=0.2: 20% of training resets are forced to |theta|=120-135 deg, |theta_dot|<=1.
- Hard replay p=0.2: 20% of each replay update batch is sampled from transitions in that same hard-start band.
- Near-down task: task-stability on evaluation starts with |theta|>=150 deg.

Result

- Hard reset p=0.2 helps 50k task-stability slightly, but reference success falls sharply.
- Hard replay p=0.2 is worse on both task-stability and reference success than ordinary full 50k.

Next Experiments

Move the ablations to representative budget

1. Ablate SimbaV2 at 100k

Experiment: remove one component at a time: feature normalization, weight projection, distributional critic, reward scaling, and official SAC recipe.

Decision: identify which change actually drives the reliability gain, using seed-level intervals and exact-grid maps.

2. Push Pendulum toward 0.99

Experiment: try plasticity and reliability fixes: ReDo, Sample Weight Decay, Fisher-guided selective forgetting, and regret-weighted auxiliary losses.

Decision: keep only changes that improve reference success, task-stability, and near-down starts without damaging critic health.

3. Move to CartPole-Swingup

Experiment: reuse the exact-grid/replay/critic-health protocol on CartPole-Swingup after the Pendulum frontier is stable.

Decision: test whether the Pendulum diagnosis transfers when exploration is genuinely harder.

Backup: Pendulum Task

Angles, observation, action, and return

Speaker Q | Q&A

State convention

- `theta = 0`: pendulum upright.
- `theta = +/- pi`: pendulum hanging down; the two ends are the same physical angle.
- Plots show `theta` in degrees: `0 deg` upright, `+/-180 deg` downward.
- Policy observation is `[cos(theta), sin(theta), theta_dot]`, not raw theta.

Return and controls

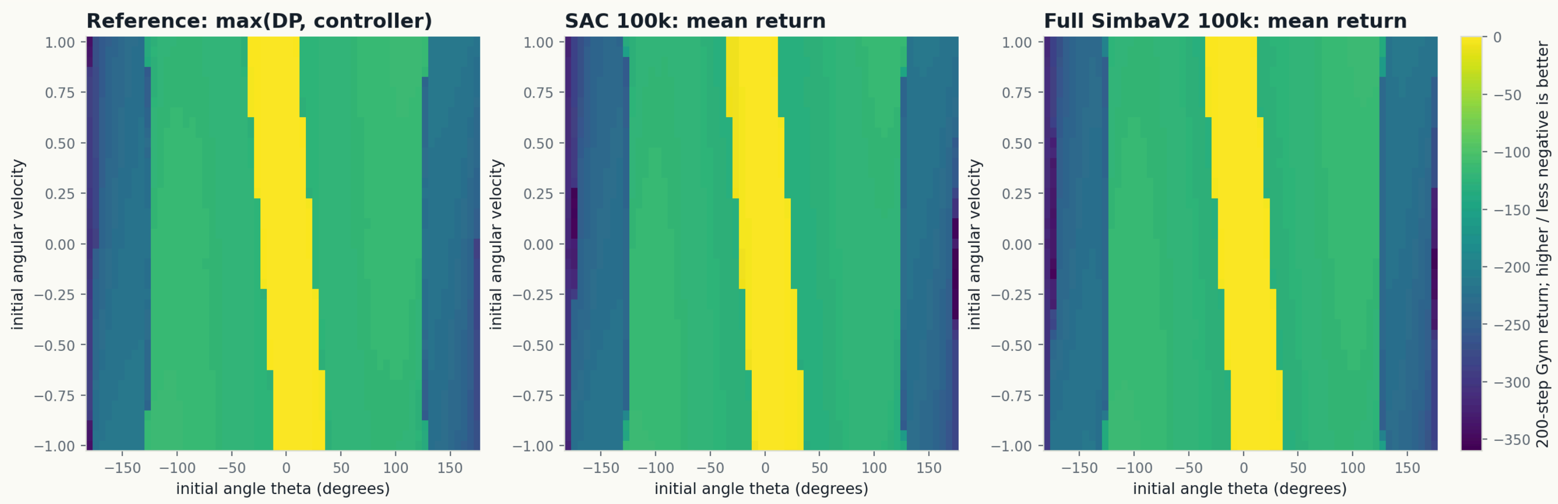
$$r_t = -\left(\mathrm{angle\_normalize}(\theta_t)^2 + 0.1\dot{\theta}_t^2 + 0.001u_t^2\right)$$

$$R = \sum_{t=0}^{199} r_t, \qquad u_t \in [-2, 2], \qquad \dot{\theta}_t \in [-8, 8]$$

- Higher return is better; the best possible per-step reward is near 0.
- The return metric rewards being upright, slow, and using little torque.
- Task-stability is separate: near upright means `cos(theta) >= 0.95` and `|theta_dot| <= 1.0`.

Backup: Raw Return Maps

Absolute returns behind the binary success maps

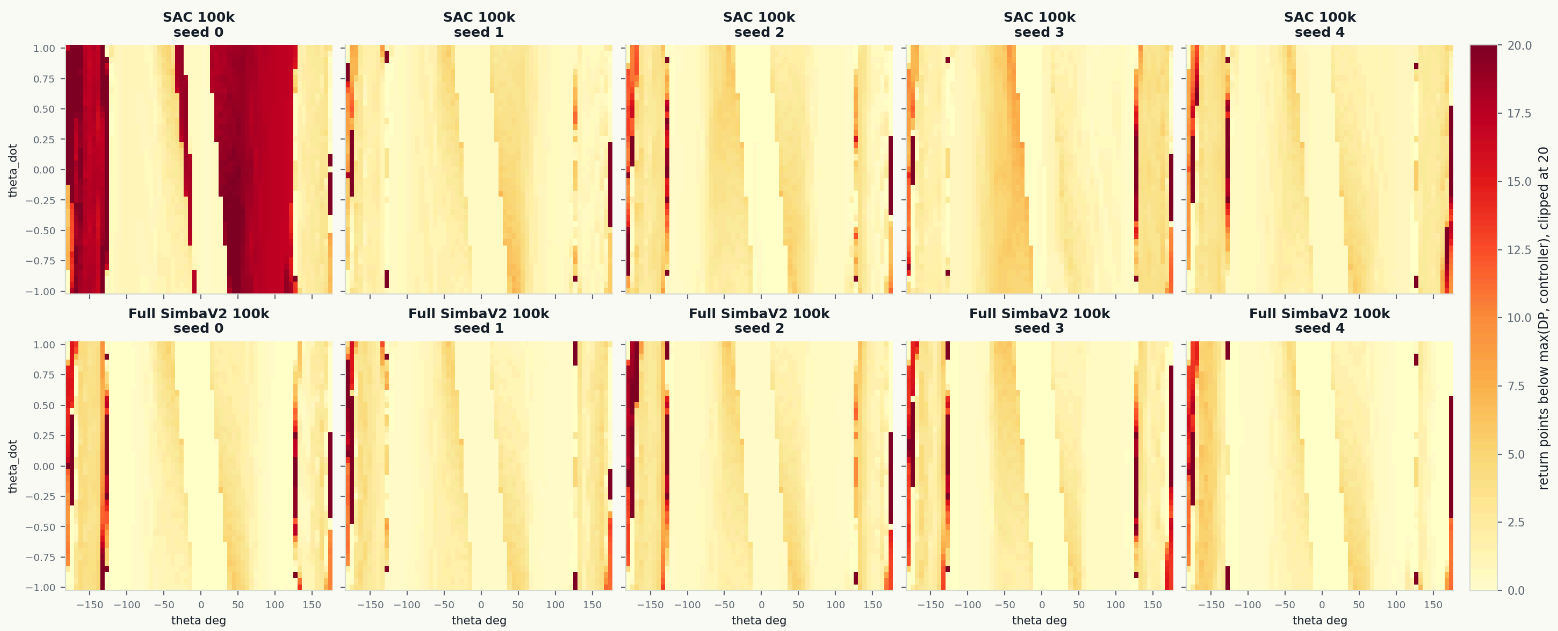


All three panels use the same color scale. The reference is `max(DP, controller)`; the learned-policy panels are mean return over five training seeds.

Backup: Per-Seed Shortfall Maps

Speaker Q | Q&A

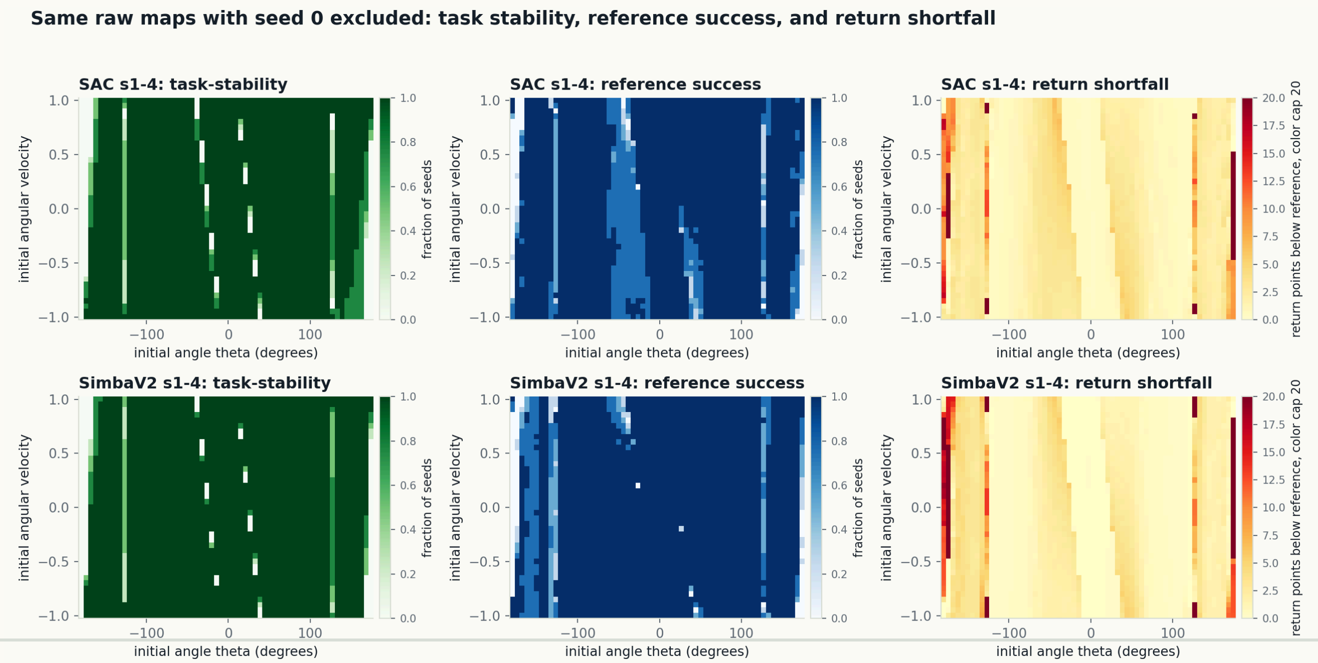
Each seed's return gap to max(DP, controller)



Each panel is one training seed. Color is return points below `max(DP, controller)`, clipped at 20; white/yellow near zero means the seed is close to the same-start reference.

Backup: Raw Plots Without Seed 0

Same plots after removing the outlier SAC seed



This is the same visualization as the main Raw Plots slide, recomputed from per-seed rollouts after excluding seed 0 for both SAC and SimbaV2. It is a sensitivity check, not the main result.

Backup: DP Reference

What DP means in this project

Speaker Q | Q&A

Dynamic programming setup

- Uses the known Gymnasium Pendulum-v1 transition equation and reward.
- No neural model is learned; Bellman backups are computed on a grid.
- Horizon 200, torque limit 2.0, velocity limit 8.0.
- State grid: 241 theta x 161 theta_dot; action grid: 81 torques.

Why it is only an approximate reference

- Continuous state and action are discretized; next-state values use bilinear interpolation.
- It is finite-horizon and optimizes Gym return, not the task-stability predicate directly.
- DP is better on 2403/2501 cells; the controller is better on 98/2501 cells.
- That is why reference success uses max(DP, controller), and task-stability feasibility is reported separately.

Backup: Hand Controller

Energy shaping plus a local PD switch

Speaker Q | Q&A

Control law used

$$E(\theta, \dot{\theta}) = \frac{1}{2}\dot{\theta}^2 + E_u \cos\theta, \qquad E_u = \frac{3g}{2l}$$

$$u = \mathrm{clip}_{[-2,2]}(-k_p\theta - k_d\dot{\theta}) \quad \text{if } |\theta| \leq 0.4,\ |\dot{\theta}| \leq 3$$

$$u = \mathrm{clip}_{[-2,2]}(-k_E(E - E_u)\dot{\theta} - 0.5\ \mathrm{sign}(\theta)1_{|\dot{\theta}|<0.1}) \quad \text{otherwise}$$

Gains: k_E=2, k_p=9, k_d=3. Implemented in `reference.py`.

Why it is in the reference

- Energy-shaping swing-up follows the classic Astrom-Furuta energy-control idea.
- The PD branch is our local stabilizer near the upright equilibrium.
- Reference success uses `max(DP, controller)` from the same initial state.
- The controller is better than DP on 98/2501 grid cells.
- It does not certify impossibility when it fails.

Citation: K. J. Astrom and K. Furuta, "Swinging up a pendulum by energy control," Automatica, 36(2), 287-295, 2000.

Backup: Bad SAC Seed

Why the SAC 100k interval is so wide

Speaker Q | Q&A

What happened

SAC 100k seed	reference	task	replay near
seed 0	52.4%	45.8%	82.5%
seed 1	93.0%	92.2%	82.4%
seed 2	92.2%	89.3%	83.0%
seed 3	85.0%	87.2%	82.4%
seed 4	92.6%	91.4%	82.8%

- Seed 0 remains the failure mode that keeps SAC's seed interval large: reference-success CI half-width is 21.6 pp.
- Replay near-upright coverage is tied, so this is not simply "did not see upright states."

Why we kept it

- Configs differ only by seed; the run completed 100k steps and 99k updates.
- Checkpoint has no NaNs/Infs; deterministic grid evaluation matched scalar Gym rollout on checked states.
- Failures are broad and mostly moderate: many cells are 10-20 return points below max(DP, controller)-5.
- Interpretation: a real optimization/reliability failure, not an obvious logging or evaluator bug.